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MILITARY SPECIFICATION

TRUCKS, LIFT, FORK, REACHING AND TIERING, NARROW AISLE, ELECTRIC, 3000 (1360 KG) AND 4000 POUNDS (1820 KG), CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers trucks, lift, fork, reaching and tiering, narrow aisle, standup rider, end control, battery powered and telescoping mast.

1.2 Classification. Trucks shall be of the following types, styles, and sizes, as specified: (See 6.2).

Type E -- Non spark-enclosed.
Type EE -- Spark-enclosed.
Style I -- Outriggers, pallet straddling.
Style II -- Outriggers, non pallet straddling.
Size 1 -- 144 inches (3660mm) lift, 95.5 inches (2420mm) overall height
Size 2 -- 130 inches (3300mm) lift, 89.5 inches (2270mm) overall height
Size 3 -- 180 inches (4570mm) lift, 95.5 inches (2420mm) overall height
Size 4 -- 205 inches (5200mm) lift, 89.5 inches (2270mm) overall height

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Navy Ships Parts Control Center, Post Office Box 2020, Attention: Code 0361, Mechanicsburg, Pa. 17055 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3930

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2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation, (see 6.2).

SPECIFICATIONS

Federal

QQ-P-416 --- Plating, Cadmium (Electrodeposited)

GG-P-455 --- Plates and Foils, Photographic (Photosensitive Anodized Aluminum)

TT-E-489 --- Enamel, Alkyd, Gloss (for Exterior or Interior Surfaces)

Military

MIL-P-514 --- Plate, Identification Instruction and Marking Blank.

MIL-G-3859 -- Grease Gun, Hand, High Pressure, Lever-Operated Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity).

MIL-M-3971 -- Meter, Time Totalizing, Operational Use, Electric.

STANDARDS

Federal

FED-STD-H28 -- Screw Thread Standards for Federal Services

FED-STD-595 -- Colors.

W-B-133 ------ Battery, Storage (Lead-Acid Industrial For Cycle Service).

Military

MIL-STD-129 -- Marking for Shipment and Storage.

MIL-STD-130 -- Identification, Marking of U. S. Military Property.

MIL-STD-162 -- Preparation for Delivery of Warehouse Materials Handling Equipment for Domestic and Overseas Shipment and Storage.

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MIL-STD-1472-- Human Engineering Design Criteria For Military Systems, Equipment and Facilities

MS-15367-------Battery, Storage, For Electrical Powered Industrial Trucks and Tractors.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (Attn: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099)

2.2 Non-Government Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B 56.1 - Safety Standard for Low Lift and High Lift Trucks

(Application for copies should be addressed to: American National Standards Institute,1430 Broadway, New York, New York 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 633 - Electrodeposited Coatings of Zinc on Iron and Steel

(Application for copies should be addressed to: American Society for Testing and Materials, 1961 Race Street, Philadelphia, PA 19103

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J356 - Welded Flash Controlled Low Carbon Steel Tubing Normalized for Bending Double Flaring, and Bending

SAE J514 - Hydraulic Tube Fitting

SAE J517 - Hydraulic Hose

SAE J518 - Hydraulic Flanged Tube, Pipe and Hose Connections, 4-Bolt Split Flange Type

SAE J524 - Seamless Low Carbon Steel Tubing Annealed for Bending and Flaring

SAE J525 - Welded and Cold Drawn Low Carbon Steel Tubing Annealed for Bending and Flaring

SAE J534 - Lubrication Fittings

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SAE J551 - Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (20-100 Mhz

SAE J925 - Minimum Access Dimensions for Construction and Industrial Machinery

(Application for copies should be addressed to: Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, Pennsylvania 15096).

UNDERWRITERS' LABORATORIES INC.

UL 583 - Standards for Safety for Electric Battery Powered Industrial Trucks

(Application for copies should be addressed to: Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, New York 11746, 333 Pfingston Road, Northbrook, IL 60062 or 1655 Scott Boulevard, Santa Clara, California 95050).

2.3 Order of Precedence In the event of a conflict between the text of this specification and the reference cited herein(except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this document supercedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. Trucks covered by this specification shall be complete and ready for operation when delivered. Batteries shall be as specified in Trucks shall be constructed and equipped to meet all requirements specified in 3.7.2.1. herein. The end item and major and minor assemblies shall be new and of a standard design or model product used by the end item manufacturer current at time of bidding, except for such minor modifications from manufacturer's standard as may be required to conform to these specification requirements. The component parts of the unit need not be the products of the same manufacturer. However, the manufacturer shall be so established in the industry that prompt and continuing service and delivery of repair parts will be assured.

3.2 First Article . When specified (see 6.2), a first produced truck shall be subjected to a first article inspection (see 6.2) in accordance with 4.3.

3.3 Material. Material not definitely specified shall be at least equal to the end item manufacturer's normal commercial product, and shall meet the test requirements specified in section 4. Reclaimed and/or recycled material shall not be used, since the intended use of the truck will be jeopardized.

3.4 Cold Storage Conditioning. When specified (see 6.2) cold storage conditioning is required to compensate for the corrosive effects resultant from condensation of moisture present in most cold storage applications. The requirements are for trucks operating in an environment where the temperature

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fluctuates from normal warehouse ambient temperature to -40°F (-40°C) or for continued operation at temperatures down to -40°F (-40°C). Sealed bearings shall be used on all pulleys, rollers, and shafts. All lubricants and oils shall be suitable for the intended service as recommended by suppliers. All electrical components subject to water shall be sprayed with an appropriate protective coating. Motors and electrical controls shall be suitable for operation in the specified environment and modified if required to eliminate condensation of moisture and possible corrosion damage.

3.5 Design. Design of trucks covered by this specification and all components therein, shall conform to accepted design practices and shall insure safe, efficient, and economical operation under normal service conditions. Where used in this specification, "easily accessible" shall refer to the truck after the battery is removed. There shall be no evidence of accelerated wear or failures or permanent deformation or leakage of fluids when operated under normal service conditions and when tested as specified herein.

3.5.1 Safety. Trucks shall conform to the applicable requirements of American National Standards Institute Publications: Standard for Electric-Battery-Powered Industrial Trucks UL 583 and Safety Standard for low lift and high lift trucks ANSI/ASME B56.1. Trucks shall comply with the requirements of OSHA 29 CFR, Chapter XVII Part 1910.

3.5.1.1 Certification. Trucks shall bear UL or other nationally recognized testing laboratory stamp of approval for type E or type EE as specified.

3.5.2 Capacity. The load capacity of the truck shall be not less than that spcified at 24-inch (610mm) load center measured from the vertical face of the forks at maximum fork height with the reach mechanism fully extended.

3.6 Electrical equipment.

3.6.1 Resistors. Resistors shall not be used in series with the motor armature for speed control. Speed shall be controlled by means of a solid-state control system as specified in 3.7.1.2. Low resistance shunts across field coils may be used to increase top speed of the motor. Such resistors as are necessary in the control circuit network shall be continuous-duty types and of a power rating sufficient to withstand all tests required by this specification.

3.6.2 Overcurrent devices. As required by UL 583, overcurrent devices shall be provided to protect the power circuits from damage resulting from internal or external wiring defects or other faulty operation. The overcurrent device(s) shall not function while performing test operations specified herein, including ramp climbing. Overcurrent devices shall be provided for each power and each control circuit. Power circuits shall be fused in the positive circuits.

3.6.3 Control panel. The control panel shall be designed for easy removal as a unit or for front removal of individual major components mounted thereon. The panel cover shall be readily removable without interference from the truck structure or other major components. Bolts, mounting studs, and nuts shall be recessed in back of panel and shall be secured so that they will not turn or

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become loose when the electrical assembly is removed or replaced on the panel. Leads to and from the panels or components shall, to the maximum extent practicable, be harnessed in compact groupings. All leads and harnesses shall be of sufficient length for ease of removal of panel or components mounted thereon. Panels shall be provided with 0.5 inch (12mm) wide rubber gasketing.

3.6.4 Enclosures and shields. All electrical components except controls and motors shall be located in panels. All cables, wiring and controls which pierce the panel shall be provided with grommets.

3.6.5 Conductors. Conductors shall be located within the truck frame and body, in conjunction with suitably designed noncombustible shielding, to provide mechanical protection from damage. All wires and cables shall be color or number coded for identification and the identification scheme shall be included in the electrical diagram. Wiring external to the truck or when maneuvering the truck shall be encased in rigid metal conduit, except that flexible metal conduit may be used at flexure areas. Similar protection shall be provided for exposed wiring runs in operator's platform area. Conductors passing through compartment walls or other metal barriers shall be protected by loom, rubber bushings, or other protective material and shall be securely anchored to prevent movement of conductors relative to the opening. All terminal connectors shall be fastened with spring lock washers, lock nuts, or other means of preventing accidental loosening as a result of vibration or unintentional movement of wire terminal ends.

3.7 Power system.

3.7.1 Motor and motor accessories.

3.7.1.1 Motors. Drive motors shall be either series or compound wound, heavy duty, industrial type, with a high starting torque, designed for efficient operation at 24 volts direct current. Motors shall be capable of withstanding the mechanical stresses and current loads induced by tests as required in section 4. The resultant strains and temperature rise shall not cause failure of any component part, excessive pitting, or other than normal blackening of the commutators. The hydraulic pump motor(s) shall have sufficient capacity to lift rated load at speeds specified herein, in addition to extension and retraction requirements. Motor(s) shall be directly connected to the pump. Motors shall be capable to withstanding sustained operation without evidence of insulation breakdown, short circuit, or failure of any part or component.

3.7.1.2 Speed controller. Speed and directional control(s) shall be provided by a vertical or rotary control(s) mounted for right hand operation. Control(s) shall automatically return to a neutral position under any of the following conditions: (a) release of the control, (b) when operator dismounts truck. The speed controller shall provide infinitely variable truck speed up to 75% - 85% of the maximum truck speed. Control and power circuits shall incorporate solid-state components that will insure operations as specified herein and minimize power losses through devices as a consequence of voltage drops through the rectifying or control elements. Control system design shall insure continuous-duty operation without excessive heating of the motors due to eddy-currents or hysteresis losses caused by current pulsations or so-called "spikes" in the applied voltage when tested as specified herein.

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3.7.1.3 Contactor(s) assembly. Contactors shall be the magnetic clapper or internal solenoid type, designed for efficient operation at 24 volts. Contactors on a given truck shall be designed for maximum interchangeability of parts with easily replaceable tips of silver or copper alloy composition and shall be manufacturer's standard component. All contactors shall be readily accessible.

3.7.2 Battery and battery accessories.

3.7.2.1 Battery. The contractor shall furnish the battery installed in the truck. Batteries shall conform to W-B-133 and MS-15367. Battery for 3000 pound (1360kg) truck shall be 24 volt, 12 cell, 11 plate, 800 amp-hour at 6 hour rate; battery for 4000 pound (1820kg) truck shall be 24 volt, 12 cell, 13 plate, 870 amp-hour at 6 hour rate, type E or EE as specified.

3.7.2.1.1 Battery connectors. Trucks shall be equipped with a 2-contact type EC standard battery connector or type SB connector with adapter furnished for connecting with mating parts of type EC connectors. Charging connectors or receptacles shall be so arranged that they do not permit movement of the vehicle under its own power while connected to the charging source. Connectors shall be color coded red.

3.7.2.1.2 Emergency battery disconnect. A quick disconnect handle shall be furnished and installed on the battery connector which is properly positioned for quick access by the operator, to disconnect all power from the battery. The device shall be conveniently located so that it can be actuated by the operator without leaving the compartment.

3.7.2.2 Battery compartment. Battery compartment or support area shall accomodate the battery specified with clearance on all sides conforming to UL 583 requirements. Effective means shall be provided to restrain the battery from moving more than a total of 1/2 inch (12mm) in a horizontal direction. The compartment shall be constructed to permit easy removal or servicing of the battery such as checking water level and refilling. When compartment construction only permits side removal of the battery, rollers or other means shall be provided to facilitate battery removal or replacement by a force not exceeding 75 pounds (330 N) to maintain movement after battery is in motion. Both sides of the compartment shall be hinged or easily detachable with hand-operable quick opening fasteners to permit battery removal, inspection, and servicing. The inside of the compartment shall be protected with an acid resistant paint to resist deterioration through normal contact with electrolyte. Compartment shall be easily accessible for cleaning and battery servicing and shall contain no inaccessible pockets or reservoirs which are difficult to keep clean. Where battery leads penetrate the battery compartment, a rubber or plastic grommet with retainer shall be installed to prevent abrasion or chaffing of the battery leads. If cover is installed, cover shall be hinged with latch and shall provide at least 90 degrees opening from the closed position.

3.7.2.3 Battery Protector. Battery discharge indicator shall be a solid state device requiring no adjustment except a perodic light bulb check or replacement. It shall be designed to protect the battery from approaching a dangerously low state of discharge by first warning the operator by means of a

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signal light, or fuel gage and signal light. After a short interval of time or after consumption of an additional amount of electrical energy, it shall interrupt the lift control circuit making the lift mechanism inoperative until the battery has been replaced or recharged. The circuit shall be such that the operator cannot activate the lift control circuit by unplugging and replugging the battery after it has been 80% discharged, (typically 1150 specific gravity). Unplugging and replugging the battery prior to reaching 80% discharge shall not reactivate the lift control circuit. The discharge indicator shall reset when a battery charged to 56% of full capacity(typically 1200 specific gravity) is put on the truck. The discharge indicator may have controls to adjust the discharge setting and the reset level to values other than 80% and 56%, provided such settings can be made without electrical test equipment and the positions of the settings are indicated by indelible markings on the controls. When a gage is used, it shall be mounted on the driver panel, and it shall display energy in the familiar empty to full type display used in gasoline vechiles. There shall be a defined point on the gage where the warning light goes on and a defined point where the truck lift mechanism becomes inoperative. Location and method of mounting shall provide plain visibility of the indicator to the driver without creating interference with truck operation, maintenance, and entry and egress movements of the operator.

3.7.3 Drive unit. The drive shall be an all gear unit with output connected to the drive axle(s). The motor, reduction unit, and drive wheel(s) shall be integrally turret mounted and shall revolve on large diameter hardened ground bearings. Gear unit, shall be totally enclosed and operate in a constant oil bath. Final reduction may be in a totally enclosed casing or be arranged for positive periodic lubrication of final drive gearing to wheel. Drive gear housings shall be equipped with filler and magnetic drain plugs readily accessible for servicing without removing other components.

3.7.4 Hydraulic system. The hydraulic system shall consist of all the hydraulic components necessary for operation of the forks and mast. The power used for the system shall be furnished by a hydraulic pump(s) directly driven by an electric motor(s) independent of the traction motor. The lifting system shall provide for smooth lowering of load at a speed not greater than 100 feet (30 meters) per minute in event of failure or damage to hydraulic lines. Provision shall be made to protect hydraulic system against excessive pressures due to overloading. All hydraulic lines shall be routed to provide the shortest line, minimum number of bends, ease of maintenance and maximum protection. Provisions shall be made to prevent damage of lines due to chafing. Sump tank and hydraulic system shall be capable of being drained without spillage. The storage tank breather and filler cap, lift, tilt, and extension cylinders shall be provided with means to protect against contamination of hydraulic fluid by direct impingement or seepage of water. A filter type breather cap shall be provided with a removable strainer not coarser than 50 mesh, installed for ease of inspection and cleaning to prevent entry of foreign material into storage tank and the hydraulic system. In lieu of the strainer, a 50 mesh accessible/cleanable screen located between the pump and the tank may be installed. A removable 25-micron, screw in type, full-flow filter with emergency bypass shall be provided to filter all the oil. The storage tank shall be of welded steel construction and shall have a capacity not less than 110 percent of the amount of fluid required for the hydralic system's operation. In addition, it shall have sufficient capacity to prevent air from entering the hydraulic system with all hydraulic pistons fully

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extended and shall have sufficient free air capacity to prevent oil from being discharged through tank filler cap when maximum return flow of oil is surged into the tank from the system. Piston rods for hydraulic cylinders shall be provided with a hard chrome plate finish. Surfaces shall be thoroughly cleaned prior to plating in accordance with normal commercial practice. Plating shall be electro-deposited hard chromium or electroless nickel finish, having uniform thickness of not less than 0.0005 inch. Lifting and other cylinders, if any, shall be provided with "wiper rings" to preclude entry of dust and dirt into the packing. Hydraulic valves and cylinders shall have means for preventing oil seepage from dripping or running onto electrical elements.

3.7.4.1 Hose reels. When hydraulic hose reels or other types of takeups are used on the mast to contain hydraulic hose, they shall be designed to facilitate containment, release and retraction of the hose without entanglement. Sheave wheels, guide rollers, guide blocks, or a mechanism in the hose reel may be used to maintain hose tension.

3.7.4.2 Hydraulic pressure hose. Hydraulic pressure hose shall conform to SAE J517, except 100R7 type hose pressure tubing shall be in accordance with SAE J524, SAE J525, or SAE J356. Tube fittings shall conform to SAE J514 or SAE J518. Bending of hose line across take-up sheave reels shall be excepted from this requirement. Suction return and bleed lines shall be SAE J517, 100R4 or oil resistant hose.

3.8 Structure.

3.8.1 Chassis and frame. The frame and related structure shall be fabricated from structural steel and steel castings at least equal to the size and design of the manufacturer's commercial frame members. The frame and related structure shall be sufficiently strong and well braced to prevent racking or twisting that might result in permanent distortion of any component part or member. The frame and related structure shall have sufficient depth to afford protection for working parts. It shall withstand the frame racking test specified in 4.5.2.12 without permanent deformation or distortion.

3.8.2 Body. The truck body shall be constructed of heavy sheet steel or plate equal to that used in the manufacturer's commercial product. In addition, the truck body shall provide protection for the drive unit assembly, battery assembly, steering mechanism, and other parts requiring protection. Provision shall be made for immediate and easy access to all working parts requiring inspection and maintenance. Covers shall be secured with bolts or quick opening fasteners removable with ordinary hand tools. Overall truck width shall be such that the truck can perform all applicable tests specified herein.

3.8.2.1 Operator's platform. The operator's platform shall be fabricated from steel plate and shall be provided with non-slip rubber tread not less than 1/4 inch (7mm) thick, cement bonded thereon. The operator's platform shall be to the right of the drive turret and shall have sufficient depth to permit the operator to stand within the plan outline of the truck and permit unobstructed egress from the rear of the truck. It shall be strong enough to prevent collapsing and pinching the operator in event of collision. The operator's compartment shall provide protective space with minimum dimensions of 14 inches (360mm) front to back and a minimum of 13 1/2 inches (340mm) width at the

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platform floor. Guards shall be constructed to withstand a compression load equal to the weight of the loaded vehicle applied in the direction of travel. If distance from floor to operator's platform exceeds 13 inches (330mm), an auxiliary step shall be provided approximately halfway between the platform and floor level. Step shall not extend beyond truck profile. Operator platform shall conform to MIL-STD-1472 using 5th percentile body dimensions for ground troops, regarding location, design, function, construction, design, visibility and operator safety.

3.8.3 Uprights. Uprights shall be of the telescopic type with sufficient strength and rigidity to withstand 300-percent of rated load with forks retracted and 200-percent with forks extended without permanent deformation or failure. Cross members of the mast structure, when in a lowered position, shall not interfere with the operator's horizontal or downward line of sight directly to the front of the vehicle. The operator shall be able to see the tip of at least one fork at any lift height without leaving his operating position when there is no load on the forks. Mast rollers with anti-friction bearings shall be provided between telescopic sections and main frame upright. Adequate means by either replacement or adjustment shall be provided in the uprights to compensate for wear between lateral rolling contacts without disassembly of the mast. Upper and lower limit stops shall be provided to prevent overtravel. The upright lift arrangement shall be designed to control premature elevation of an intermediate section during raising or "hanging-up" of these sections during lowering.

3.8.3.1 Forks. Steel forks shall be provided which shall be capable of withstanding, without permanent deformation, a static test load of 300-percent of rated load as well as stresses induced by tests specified in section 4. Forks shall be attached in a manner which prevents lateral slippage or permanent deformation of lift carriage components. Forks shall be adjustable laterally without the need for hand tools. Taper shall be on the underside of the fork. Fork length shall be as specified. Fork mounting shall be of the shaft type. Tolerance for length shall be +0 -1/2 inches (+0 -12mm). Unless otherwise specified, additional details are as follows:

TABLE I. Fork Dimensions.

Thickness (maximum)	2.0625 inches (52mm)
Thickness at tip	0.5 inches (12mm)
Length of taper to tip of fork	23 inches max (360mm)
Width (maximum)	5.0625 inches (130mm)
Length	40 inches (107mm)
Spacing, minimum, outside (forks spread)	28.5 inches (720mm)
Spacing, maximum, outside (forks together)	18 inches (460mm)

3.8.3.2 Load back rest. A metal back rest removable without disturbing any other component shall be provided. It shall be 47 to 48 inches

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(1190-1220mm) high measured from load carrying surfaces of the forks and shall be a minimum width at least equal to the width of the carriage. It shall have width, capacity and strength sufficient to prevent a rated load 48 inches (1220mm) high or less from forcing back against the uprights. The backrest shall be firmly attached to the lift carriage but shall be readily removable with common hand tools. The back rest shall not pinch the load when the forks are tilted backwards. Backrest design shall not unduly obstruct operator's vision. The spacing of vertical members shall be no more than 6 inches (150mm) between members.

3.8.3.3 Fork carriage. The fork carriage shall provide lateral and longitudinal absorption of load thrusts without failure of any component when subjected to 300-percent of rated load. The carriage shall be mounted on the mast on adjustable rollers equipped with antifriction bearings.

3.8.4 Outriggers. Outriggers shall be fabricated from steel plate or bar steel formed to shape as an integral part of the frame to provide a rigid structure and shall be capable of withstanding without permanent deformation all stresses induced by tests specified in section 4.

3.8.4.1 Style I trucks. Inside spacing between outriggers shall be not less than 50 inches (1270mm) to permit straddling of a 40-inch (1020mm)by 48-inch (1220mm) pallet on the forks at floor level. Outriggers shall have an outside width not exceeding 63 inches (1520mm).

3.8.4.2 Style II trucks. Outriggers for style II trucks shall have an outside width of 43 inches (1090mm) + 0.5 inch (12mm) maximum.

3.8.4.3 Reach mechanism. Reach mechanism shall permit extension of the forks or fork assembly to a point where the front vertical face of the forks is at least flush with and extend at least 24 inches (100mm) in front of the forward edge of the load wheels, tires, and outriggers to permit handling of loads at floor level that are off center laterally or would otherwise conflict with the outriggers or load wheels. The reach mechanism structure shall be capable of meeting the performance requirements specified herein and shall be sufficiently strong and well braced to prevent racking or twisting that might result in permanent distortion of any component part. The design shall be particularly well suited to withstand stresses which result from bumping of one fork on one corner of the load or against an obstruction. Antifriction bearings, bronze, or steel bushing shall be provided for the rollers reach mechanism.

3.8.5 Wheels. Trucks shall be mounted on flat-faced steel or malleable iron wheels, mounted on ball or tapered roller bearings and equipped with one of the following configurations: (1) two independently suspended and driven drive wheels; (2) one drive wheel and one mechanically linked idler wheel; or (3) one drive wheel and one or more caster follower wheels. The non-drive wheels shall be loaded or the entire system designed for laterally articulated mounting to maintain drive wheel contact with uneven floors. Trucks shall have two articulated load wheels in each outrigger. Wheels, when equipped with new tires, shall clear any part of the truck structure by at least 0.25 inch (6mm) under all conditions of operation. Readily accessible grease fittings or pre-lubricated sealed bearings shall be furnished to provide lubrication in accordance with good automotive practice. Load wheels on outriggers shall be

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provided with means to assure equal load distribution on each of the load wheels with and without the rated load on the truck.

3.8.5.1 Tires. The drive wheel tires shall be industrial solid rubber, smooth low profile type or wide lug cushion type, molded onto a steel band. Band shall be of the press-on type or shall have an integral flange for direct bolting onto a driving hub. Tires for caster wheels shall be polyurethane and either the press-on or mold-on type. Minimum tire size for the drive wheel shall be 10 inches (254mm) diameter with 5 inches (130mm) face. Minimum tire size for caster or idler wheel(s) shall be 6 inches (150mm) diameter with 2 inches (50mm) face for duals and 8 inches (200mm) diameter with 3 inches (8mm) face for singles. Load wheel tires shall be 4.5 inches (120mm) diameter minimum by 2 inches (50mm) width minimum.

3.8.6 Steering. Steering shall be accomplished by a horizontal steering wheel or tiller equipped with steering knob for convenient operation. The steering control shall have the hand wheel or tiller mounted with the upper face 40 to 45 inches (1020 to 1140mm) above the operator's platform and in such a manner that will permit the operator to steer with his left hand when facing the load. The steering mechanism shall turn the drive wheel to provide for right angle stacking in one motion as specified herein. The steering mechanism shall turn the drive wheel through spur gear reduction or accepted commercial equivalent reduction device which shall permit ease of operation. A hydraulic steering booster or other power assisted steering shall be provided. Clockwise or right hand rotation of the steering wheel shall provide for right hand or clockwise turning of the truck while in forward motion (load first). Provision shall be made for precision adjustment to compensate for wear of all major steering parts and for lubrication of all friction points by accessible grease fittings or a lubricant reservoir. The hydraulic steering or power assisted steering fluid reservoir shall be readily accessible for checking of lubricant level.

3.8.7 Brakes. Brakes shall be hydraulically or mechanically activated andmechanically deactivated and of the external contracting type, disc type, or internal expanding shoe type mounted on the traction motor shaft, countershaft, or drive wheel. Brake lining, when used, shall be of molded or woven type and shall be cement-bonded or riveted to the brake shoe. Brakes shall be operated by the operator's left foot and shall be automatically spring applied when the brake is released or when the operator leaves the truck. The braking system shall be electrically interlocked with a switch or switches which shall interrupt the electrical circuit to the traction motor when the brake is applied. The brake adjustment shall provide for at least 90 percent contact of brake lining with drum throughout service life of the lining. The operator's brake pedal shall be located within a clear space at least 6 inches (150mm) wide. Pedal shall be at least 4.5 inches (110mm) wide and provided with a rubber covering or other non-slip material to prevent slippage. The brake pedal, when depressed to fully release the brake, shall be in a horizontal plane, flush with the operator's platform.

3.8.8 Sideshifter attachment. The truck shall be equipped with a side shift attachment. Side shifter shall be hydraulically controlled and shall provide a minimum movement of 2 inches (50mm) on each side of center with rated load.

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3.8.9 Truck housing. All functioning components of the truck, except controls, steering handle, wheels, mast and fork assembly shall be enclosed in a housing. The housing shall be designed to permit easy access to all working parts requiring inspection, servicing and maintenence. The height and width of the housing shall be such as to provide unobstructed view of fork tips when the operator is in the operating position.

3.9 Controls and instrumentation. Controls and instrumentation shall be moisture and weather resistant. All controls shall return to neutral upon release of the control.

3.9.1 Controls. All switch and lever type controls shall be provided with permanent position markings (decals are not acceptable) either at the switch or lever or, in the case of the directional speed controls, on a diagram visible to the operator. Distance from the near edge of one knob to the near edge of an adjacent knob or other surface shall be a minimum of 1.5 inches (40mm). All controls shall have a directional motion insofar as possible. Motion of the hand control shall be similar to the resulting motion of the truck operation to be performed.

3.9.1.1 Directional speed controls. Selective forward and reverse speed control(s) shall be provided and shall be conveniently located for safe and efficient right hand operation with operator facing load. Controls shall also provide for safe, convenient, and efficient operation when operator faces away from load.

3.9.1.2 Lift, reach and tilt controls. Lift, reach and tilt controls shall be self-centering, conveniently located for right-hand operation. Controls shall be located such that lift control is at the extreme right, reach control at the next position, and tilt at the extreme left. The controls shall provide for lifting and lowering of the load when the truck is in motion without requiring the operator to release the steering control or utilize any part of his body other than hands to operate controls. Rearward or upward motion of lift, reach, and tilt control levers shall elevate, retract or tilt load rearward. Modulation of the speed of lift, or lowering, reach, and tilt shall be readily controllable. Lifting speed shall be infinitely variable from zero to the maximum lifting speed specified. Lift, reach and tilt controls may be incorporated in a self centering multifunction control handle located for right hand operation in lieu of individual control levers.

3.9.1.3 Accelerator control. A hand operated accelerator control shall be provided. A control button or grip may be combined with the directional control and shall be installed for comfortable right hand operation.

3.9.1.4 Extension retraction control. The forks shall be extended by movement of the control from the center position to the forward position. Retraction of the forks shall be accomplished by movement of the control from the center position to the rearward position. Means shall be provided to prevent jerky stopping or starting of forks at the end of the extension and retraction movements.

3.9.1.5 Remote control. When specified (see 6.2), each truck shall be equipped with separate controls in such a position that the operator, while standing on a safety pallet placed on the forks, shall be able to operate the

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elevating and lowering mechanism over its complete range. These controls shall be so located that when not in use they will not interfere with normal operation of the truck or constitute a safety hazard. These controls shall be protected so they cannot be operated or damaged by commodities falling on them. They shall be so connected that operation of the lift mechanism cannot be accomplished unless the deadman brake is set. Interlocks shall be provided so that only up and down motion is possible, preventing another person from moving the vehicle in a horizontal position, while the operator is on the forks. Lift-lower controls shall be spring-loaded to return to the off position upon release. An automatic take-up device shall be provided on the mast or truck body to prevent interference with normal truck operation. Lift-lower controls shall be arranged for two hand operation so that operator must actuate separate controls wilth both hands silmultaneously to cause carriage to elevate or lower in order to prevent any possible accidents to either hand or arm while carriage is elevating or lowering. Means shall be provided for the operator on the platform to shut off the power of the truck. Cables to the remote control shall be heavy duty, oil resistant type and protected from damage by location and position regardless of lift. Suitable means shall be provided to guide these cables within the truck body so accidental scraping against walls and other objects will not damage the cables.

3.9.1.6 Key switch. A key operated switch and key shall be furnished (see 6.5). The key shall be removable only when the switch is in the "off" position.When the switch is in the "off" position, control circuits shall be open. All trucks procured under each contract shall be keyed identically. Two keys shall be supplied with the truck.

3.9.1.7 Sideshift control. The sideshift control shall be conveniently located to the left of the tilt control for right hand operation. Rearward motion of the sideshift control shall shift the forks to the right and forward motion shall shift the forks to the left. Side shift control may be incorporated into the lift, tilt, and reach control.

3.9.2 Instruments. All gages and meters shall be flush mounted in a location convenient to and plainly visible to the operator standing on the operator's platform.

3.9.2.1 Hour meter. An electric operational hour meter which registers the total number of operating hours of the traction and pump motors only shall be provided. Hour meters shall conform to Specification MIL-M-3971, type I, grade B or of the quartz type. Hour meter shall be wired to record only powered operations and shall operate only when deadman pedal is depressed. Hour meter shall be visible to the operator at all times.

3.10 Accessories. Where applicable, the provisions of UL 583 shall apply. All accessories shall be constructed to resist moisture, rain and weather.

3.10.1 Horn. Truck shall be equipped with an electric push button operated horn. Push button shall be mounted and located for convenient operation by the operator. The horn shall be operable without requiring the operator to release the steering or directional controls.

3.10.2 Operator's overhead safety guard. An overhead safety guard for the protection of the operator shall be provided in accordance with ANSI B56.1.

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Guard height, measured from ground level to top side of the guard frame members shall not exceed 95.5 (2420mm) inches for sizes 1 and 3 and 89.5 inches (2270mm) for size 2 and 4. The guard shall extend to within 2.5 inches (64mm) of the rear plan outline of the truck. Guard design shall not interfere with operation of truck, or with normal movements of the operator when entering, leaving, or operating the truck. Structural steel supports (2) shall run from the back of the overhead guard to the truck frame to protect the operator.

3.10.3 Safety shield. A transparent tempered or laminated safety glass or transparent plastic shield 0.5 inches (12mm) minimum thickness or combination of plastic sheet/safety glass and screening shall be mounted on the mast in such a manner as to prevent injury to the operator's hands or arms by the lift chains or cross members. The shield shall not interfere with the operational requirements of the truck or with the control knob clearances. It shall not obstruct the vision of the operator.

3.10.4 Floodlight. Each truck shall be equipped with one hand adjustable, sealed beam floodlight of not less than 25 watts, mounted on the mast and located for right hand operation. Minimum adjustment shall be 45 degrees above and below the horizontal plane and 150 degrees clockwise and 150 degrees counterclockwise from the front in the vertical plane. Adequate protection by position or guard shall be provided to prevent damage to the light under all operating conditions. An on-off switch shall be incorporated into the control panel or on the light if within reach of the operator. Floodlight shall be wired independent of other electrical circuits.

3.11 Performance

3.11.1 Lifting speed. Speed of lift with rated load. Shall be not less 28 feet (8.5m) per minute for 3000 pound (1360 kg) type E truck and 24 feet (7m) per minute for 4000 pound (1820 kg) type E truck, not less than 21 feet (6m) per minute for 3000 pound (1360 kg) type EE truck and 18 feet (5.5m) per minute for 4000 pound (1820 kg) type EE truck from floor to maximum fork height and not less than 40 feet (12m) per minute without load under the same conditions, forks retracted in both instances.

3.11.2 Lowering speed. Speed of lowering the unloaded forks (ambient temperature between 65 degrees F and 85 degrees F (18 degrees and 29 degrees C)), forks fully retracted, shall be not less than 40 feet (12m) per minute or more than 100 feet (30m) per minute. The hydraulic control system shall be damped to reduce shock and prevent the truck from overturning when any load is lowered and stopped suddenly.

3.11.3 Extension and retraction. Truck shall be capable of fully extending and retracting rated load at any height in not more than 6 seconds.

3.11.4 Steering. The drive assembly shall be capable of being turned from extreme right to extreme left and vice versa in not more than 6 turns of the hand steering wheel with a maximum steering effort of 5 pounds (22N) when the truck is stationary on a smooth dry concrete floor with and without rated load in the carrying position, forks retracted. The steering gear assembly shall be capable of withstanding an unbalanced tangential force of 25 pounds (110N) applied to the steering wheel without permanently distorting or deforming any part of the steering mechanism. Full time power steering is required.

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3.11.5 Speed. When carrying a rated load on a level surface, trucks shall be capable of forward and rearward speeds of not less than 5 miles (6.5km) per hour when equipped with a fully charged battery.

3.11.6 Slope ascension, forward direction. The truck shall be capable of ascending a 10 percent grade on a dry concrete surface when carrying the rated load. Length of ramp shall be not less than 15 feet (4.6m). The truck shall be able to accelerate from a dead stop on a 5 percent slope without load.

3.11.7 Acceleration. From a standing start, the truck carrying rated load shall have an average acceleration rate in the forward and reverse directions such as to negotiate a distance of 22 feet (7m) in not more than 7 seconds. There shall be not more than one second delay in movement of the truck when truck is accelerated from a stop.

3.11.8 Service brakes. The truck shall conform to the braking requirements of ANSI B56.1.

3.11.9 Fork tilt. The tilt mechanism shall be capable of bringing rated load at least 2 degrees forward when fully elevated and extended. It shall be capable of providing a positive rearward tilt of 4 degrees minimum with or without rated load on the forks at a load carrying position of 17 to 24 inches (430 to 610 mm) above the floor with the forks retracted. All of the above measurements are based on the relationship of the top face of the forks with the truck on a level flat platform. Forward and rearward tilt shall be controllable at all fork heights or extensions.

3.11.10 Overall height. Overall height of unloaded truck, with mast lowered, measured from the ground to the top of the highest part of the truck, shall not exceed 95.5 inches (2420mm) for sizes 1 and 3 and 89.5 inches (2270mm) for size 2 and 4.

3.11.11 Minimum fork height. Fork height of loaded truck, measured from the ground to the top face of the forks with forks horizontal, uprights vertical, and load extended shall not be less than 144 inches (3660mm) for size 1, 130 inches (3300mm) for size 2, 180 inches (4570mm) for size 3, and 205 inches (5200mm) for size 4.

3.11.12 Free lift height. Height of forks without exceeding specified collapsed mast height shall be not less than 40 inches (1016mm) with rated load on the forks for sizes 3 and 4. Sizes 1 and 2 shall have a minimum free lift of 6 inches (152mm) with rated load.

3.11.13 Underclearance. The truck shall have sufficient underclearace to permit operation without load from one horizontal plane to another, up and down a 10 percent slope. The truck shall have sufficient underclearance to permit operation with load from one horizontal plane to another, up and down a 10-percent slope. With the forks flat on the operating surface the truck shall have adequate underclearance to permit operation.

3.11.14 Hydraulic system. The hydraulic system shall show no permanent deformation or failure or leakage when the truck is subjected to 300-percent of

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rated load, forks retracted, and 200-percent of rated load, forks extended. This requirement does not include the pump. The pressure relief valve(s) shall be set to open when pressure equals 110-percent of the maximum pressure required to perform any operation supplied by a given pump(s).

3.11.15 Endurance test. The truck shall be capable of undergoing the endurance test specified in 4.5.2.19 without deformation, breakdown, failure, or replacement of any component (except for the battery), or any other defect and with no change in performance requirements specified herein and with no appreciable signs of wear in either the steering mechanism, the lift and tilt assembly, extension-retraction assembly, or drive assembly. Wear in the steering mechanism is considered to be an increase in free play of wheel or more than 1.5 inches (40mm) measured at the steering wheel outer periphery. Wear in the lift assembly is defined as any apparent inability to lift rated load to maximum lift height and hold for a period of not less than 5 minutes and with not more than one inch (25mm) drift.

3.11.16 Maneuverability. Trucks shall be capable of right angle stacking in aisles specified in Table II with a 40 inch (1020mm) by 48 inch (1220mm) loaded pallet (48 inch (1220mm) dimension across the forks). Trucks shall be capable of removing or placing pallets in a stack with listed clearance on either side of the pallet; time required for removal or placement shall be less than 60 seconds.

Table II. Maneuverability.

CAPCITY TRUCK LBS.(KG)	STYLE	SIZE	AISLE WIDTH MAX IN. (MM.)	CLEARANCE MAX. IN. (MM.)
3000 (1360)	I	1,2,3 & 4	87 (2205)	4 (100)
3000 (1360)	II	1,2	84 (2130)	3 (76)
3000 (1820)	II	3,4	88 (2230)	3 (76)
4000 (1820)	II	3,4	88 (2230)	3 (76)
4000 (1820)	II	1,2	84 (2130)	3 (76)

3.11.17 Stability. The truck shall conform to the stability requirements of ANSI B56.1.

3.11.18 Overhead guard. Overhead guard shall conform to the requirements of ANSI B56.1.

3.11.19 Power Consumption. The truck, when equipped with a fully charged battery as specified shall operate continuously for not less than 5 hours when tested as specified in 4.5.2.18.

3.11.20 Retraction overload. Forks, mast assembly and related structures shall show no permanent deformation or failure when subjected to a retraction overload of 300 percent of rated load when tested as specified in 4.5.2.10.2.

3.11.21 Extension overload. Forks, mast assembly and related structures shall show no permanent deformation or failure when subjected to an extension

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overload of 200 percent of rated load when tested as specified in 4.5.2.10.3.

3.12 Bearings. Rotating parts shall be mounted on ball or roller bearings to the extent normally provided in standard commerical practice and in accordance with the best automotive practice. Bearing load capacity, including applicable safety and life factors, as recommended by the bearing manufacturer, shall be equal to or greater than maximum bearing loading under normal truck operation.

3.13. Lubricants. The truck shall operate as specified when lubricated with standard military lubricants.

3.13.1 Lubrication fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch(250mm) flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panel and plates equipped with hand-operable, quick-disconnect fasteners or no more than eight common fasteners may be removed to provide accessibility.

3.13.2 Lubrication seals. All lubrication seals shall be properly sized and seated to prevent leakage of lubricant through or around positioned seals.

3.13.3 Screw threads. Screw threads shall be in accordance with Federal Standard FED-STD-H28.

3.13.4 Pressure release device. An automatic release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.14 Finish treatment and painting. All exterior surfaces of the truck shall be thoroughly cleaned and shall be dry and free from mill scale, oil, grease, dirt, and rust and shall be painted as soon as practicable after cleaning with not less than one coat of primer and two coats of synthetic enamel or with not less than two coats of primerless modified alkyd paint requiring no primer coat. Primer and finish coats shall be applied in accordance with normal commerical practice to assure complete coverage and durability of the finish. Unless otherwise specified (see 6.2) the finish color shall be No. 13538 (gloss yellow) in accordance with FED-STD-595A except mast and overhead guard may be painted black with white markings as required. The finish coat when dry shall be a smooth even surface, free from runs, sags, blisters, peels, chips, and areas of no film or thin film. Surfaces of components and assemblies of the truck not normally painted with a finish coat shall be cleaned and prepared in accordance with standard commercial practice for the forklift truck industry. Working surfaces (i.e. lift chains, sprockets, and rollers, etc.) plastics, instruments, and such other components not normally painted with the finish coat shall be adequately protected or masked from possible overspray.

3.14.1 Cold Storage Marking. Trucks with cold storage protection features shall be marked with a single 6 inch wide diagonal side stripe, white, (Color Number 17886 of FED-STD-595). The strip shall bisect each side of the vehicle at a 45 degree angle from lower rear of truck. In addition, trucks for cold storage use shall be identified with words COLD STORAGE USE APPROVED in black

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(Color Number 17038 of FED-STD-595) 1-1/2 inch high lock type capital characters located in a conspicuous space on each side of the truck.

3.15 Plating. Fastening devices such as cotter pins, washers, nuts and bolts required to fabricate the truck except those in contact with oils in reservoirs and those inside corrosion resistant components shall be zinc plated, cadmium plated , made of corrosion resisting material or shall be black oxide coated. Zinc plating shall be in accordance with ASTM B633, type I, SC 1 and cadmium plating shall be in accordance with QQ-P-416, type I, class 3.

3.16 Electromagnetic interference. The truck shall be equipped for control of electromagnetic interference in accordance with SAE J551c.

3.17 Identification marking. The truck shall be identified with an identification plate conforming to MIL-P-514, type I, style 1, Composition C, (GG-P-455 Type I, Grade A, Class 1) which shall be located in a visible and safe location on the instrument panel or firewall of the truck. The identifying data to be applied to the identification plates shall be as indicated herein and as specified in MIL-STD-130. Identification data shall include as a minimum contract number, serial number, model number, safety rating, empty weight, payload, location and capacity of lifting and tiedown points and center of gravity and battery data. All plates shall be securely attached to the truck with screws, bolts, or rivets and shall be furnished and mounted by the supplier.

3.18 Instruction plates. The forklift shall be equipped with metal or pressure sensitive instruction labels including warnings, cautions and diagrams describing any special or important procedures to be followed in assembly, operating or servicing the forklift.

3.19 Battery plate. Each truck shall be equipped with a battery information plate attached to the truck conforming to MIL-P-514, type III, composition C (GG-P-455,type I, grade A, class 1) material. As a minimum, the plate shall indicate battery weight, dimensions, ampere hour capacity at 6 hour rate, voltage, number of cells, number of plates, cable lead length and position.

3.20 Registration number and vehicle marking. Trucks shall be marked in black color number 17038 in accordance with Federal Standard No. 595. Marking paint shall be in accordance with TT-E-489. Characters shall be block-type capitals and Arabic numerals. Marking location and information and character size shall be as follows:

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Table III. Vehicle Marking.

Information	Example	Location	Character Size
Capacity	3,000 Lbs. Capacity	On each side of mast	2 inch height (50mm)
Gross Vehicle Weight	7,000 Lbs.	On each side	2 inch height (50mm)
Rider warning	NO RIDERS	On rear of mast	2 inch height (50mm)
Service Designation	U.S. Navy	On each side and rear	1-1/2 inch height (40mm)
Type Designation	EE	On each side and rear	4 inch height (100mm)
Registration Number	13-35674	On each side	3 inch height (80mm)

3.21 Maintainability. Provision shall be made for adjustment, servicing and replacement of all electrical assemblies and components, battery, wearing parts of lift and tilt mechanism, brakes and components, wearing parts of steering assembly, tires, wheels, lights, and horn without the removal of any major parts other than access covers. If hand access openings are used on the exterior of the truck, the edges of each opening shall be smooth and shall be provided with a hinged cover, whenever it is required. Dimensions of access openings shall be in accordance with SAE J925 for a bare hand or normal clothed man. Drainage of lubricants shall be accomplished without draining on any part of the truck. When drains are located over other parts of truck, integral tubes, pipes, troughs or other means shall be used to convey the lubricant from the drain to a container placed on the ground. Drains shall be accessible without the removal or adjustment of accessories or parts other than opening covers. Drain holes with removable drain plugs or covers shall provide for complete drainage of all reservoirs and driveline housings without disconnecting hose(s). The hydraulic system reservoir shall be equipped with a dipstick to determine the fluid level. As a minimum, the dipstick shall be graduated with two marks indicating 'full' and 'add'. The dipstick shall be accessible without the use of hand tools. The truck shall not require scheduled maintenance at intervals of less than 1,000 circuits of the endurance test as specified in 4.5.2.19.

3.22 Interchangeability. Corresponding units and replaceable assemblies, subassemblies, and parts having the same part number shall be physically and functionally interchangeable as complete units without modification. Where dimensions, ratings, tolerances, and other characteristics are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing.

3.23 Workmanship. The truck shall withstand any operation specified

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herein without permanent deformation, breakage of connections, malfunction, or component interference caused by improper workmanship. All parts of the truck before and after painting shall be clean and free from sand, rust, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. Edges and surfaces exposed to operating and maintenance personnel shall be smooth and rounded to the extent that a hazardous surface does not exist.

3.23.1 Steel and other metal fabrication. Metals used in the fabrication of the truck shall provide original quality surface finish and shall be free from kinks and sharp bends. Metals having corroded or pitted surfaces are not acceptable. The straightening of materials shall be done by methods that will not cause weakening or injury to the material. Burrs and sharp edges in holes and on sheets, plates, and members shall be removed sufficiently to assure correct fits and to prevent loosening of fasteners and damage to components. Flame cutting with a tip suitable for the thickness of metal may be employed instead of shearing or sawing. Splatter shall be removed from exposed cuts and from re-entrant cuts. Heated metals shall be allowed to cool slowly, except in the performance of designed heat treatment, and overheating shall be avoided in accordance with the recommendations of the metal manufacturer. All bends of a major character and all modular assembly fabrication shall be made with controlled means to insure uniformity of size and shape.

3.23.2 Welds. Welded joints shall be sound, smooth, free from pits, holes, fissures, and rough projecting edges and shall be thoroughly fused to the base metal.

3.23.3 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers or lockwashers or a combination of the two shall be provided on all bolts, studs, and capscrews having straight thread. Selflocking nuts are acceptable in lieu of standard nuts and lockwashers. All fasteners shall be correctly tightened and shall have full engagement.

3.23.4 Castings. Castings shall be of the best quality normally used in commercial practice and shall be of uniform quality and condition, free from blowholes, porosity, hard spots, shrinkage defects, cracks, misplaced coring, or other injurious defects. Castings shall be commercially machinable and shall not be repaired, plugged, or welded to adversely affect the strength, use, or machinability. They shall be of such quality as to preclude weeping of internal fluids through the castings as a result of porosity and to permit machining for leak free sealing at cover and closure points. Castings shall be suitably spot faced or machined to assure proper seating of fasteners such as bolts, lockwashers, nut assemblies and cap screws.

3.23.5 Rivet connections. Rivet holes, in the sizes recommended in standard practice, shall be accurately punched or drilled and shall have burrs removed. Rivets shall be driven with power tools. Rivet head shall be full, neatly made, concentric with rivet holes and in full contact with the surface of the member. Excessive upsetting of rivets to fill holes will not be acceptable.

3.23.6 Machine work. All parts shall be manufactured to gage through the use of correct jigs, fixtures, or tape controlled machines or combination of these. Like parts shall be interchangeable.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections in this specification where such inspections are deemed necessary to assure conformance of supplies and services.

4.1.1 Responsibility for compliance. All items shall meet the requirements of section 3 and 5. Inspection requirements shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all the requirements of the contract. Sampling inspection, as a part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-article inspection (see 4.3)

(b) Quality conformance inspection (see 4.4)

(c) Inspection of preparation for delivery (see 4.7)

4.3 First-produced truck inspection and tests. Truck furnished in accordance with 3.2 (When First Article Testing is required) shall be examined for the defects marked "X" in column 1 of Table IV and subjected to tests marked X in column 1 of Table V. Acceptance of a first-produced truck shall not exclude the remaining trucks from the quality conformance inspection requirements. Failure of a first-produced truck to meet any requirement shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. All trucks under contract shall be examined for defects marked "X" in column 2 of Table IV. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. All trucks under contract shall be subjected to tests marked X in column 2 of Table V. Failure to meet any test requirement shall be cause for rejection.

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4.5 Inspection procedure.

4.5.1 Examination. Truck(s) shall be examined for defects as specified in 4.3 or 4.4.1 as applicable.

4.5.2 Tests. Tests to determine compliance with the specified requirements shall be conducted as follows.

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TABLE IV Examination Schedule.

First Produced	Quality Conformance	Defects	Requirement Paragraph
1	2	3	4
X	X	Assembly not complete or not correct	Section 3
X	X	Components missing or not as specified	Section 3
X	-	Materials not as specified	3.3
X	X	Cold storage conditioning not as specified	3.4
X	-	Safety requirements not as specified for type E or Type EE truck certification	3.5.1 and 3.5.1.1
X	X	Overcurrent devices not provided with each circuit	3.6.2
X	-	Means of identification for conductors not provided	3.6.5
X	-	Vehicle not designed for 24 volt D.C. operation	3.7.1.1
X	-	Contactors for speed controller not easily replaceable	3.7.1.3
X	-	Battery connectors not as specified	3.7.2.1.1
X	X	Battery restraint not as specified	3.7.2.2
X	-	Drive assembly not all gear drive unit	3.7.3
X	X	Hydraulic system not as specified, does not functioin, or leaks	3.7.4
X	-	Contamination present in hydraulic reservoir return, and suction line	3.7.4
X	-	Piston rod not finished to 0.0005" chrome plate	3.7.4
	-	Frame depth does not protect working parts	3.8.1
X	-	Fork dimensions not as specified	3.8.3.1
X	-	Ball or tapered roller bearings not provided on wheels	3.8.5
X	X	Accessible grease fittings not provided for all rotating parts	3.8.5
X	-	Tires not as specified	3.8.5.1
X	-	Load wheels not made of polyurethane	3.8.5.1
X	-	Steering system not as specified, does not function, or leaks	3.8.6
X	-	Brakes not as specified or do not function	3.8.7
X	-	Truck body does not completely cover drive unit assembly, steering mechanism and speed controller when these components are not housed in individual containers	3.8.9
X	-	Easy access to all working parts is not provided	3.8.9
X	-	Controls not moisture and weather resistant	3.9
X	-	All controls are not provided with heavy moisture and weather resistant markings	3.9
X	X	Directional speed controls and push button requirements do not function as specified	3.9.1.1
X	-	Lifting reach, and tilt controls are not specified	3.9.1.2
X	X	Extension and retraction controls not as specified	3.9.1.4

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TABLE IV Examination Schedule, continued

First Produced	Quality Conform-ance	Defects	Requirement Paragraph
1	2	3	4
X	X	Remote control not as specified	3.9.1.5
X	X	Instruments not located as specified or do not function correctly	3.9.2
X	X	Treatment and painting not as specified	3.14
X	-	Electromagntic interference not as specified	3.16
X	X	Nameplate and special marking incorrect or not as specified	3.17 thru 3.20
X	X	Inspection of welds	3.23.2

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Table V Test Schedule

First Produced 1	Quality Conformance 2	Post Test 3	Test	Test Para.	Requir. Para.
X		X	Fork height	4.5.2.1.1	3.11.11
X		X	Free lift height	4.5.2.1.2	3.11.12
X		X	Fork tilt	4.5.2.1.3	3.11.9
X	X	X	Lifting and lowering speed	4.5.2.1.4	3.11.1 3.11.2
X	X	X	Extension and retraction	4.5.2.2	3.11.3
X	X	X	Underclearance	4.5.2.3	3.11.13
X		X	Steering gear	4.5.2.4.1	3.11.4
X		X	Steering overload	4.5.2.4.2	3.11.4
X		X	Free play	4.5.2.4.3	3.11.15
X	X	X	Speed	4.5.2.5	3.11.5
X	X	X	Acceleration	4.5.2.6	3.11.7
X	X	X	Slope ascension	4.5.2.7	3.11.6
X		X	Maneuverability	4.5.2.8	3.11.16
X			Load backrest	4.5.2.9	3.8.3.2
X			Shock	4.5.2.10.1	
X			Retraction overload	4.5.2.10.2	3.11.20
X			Extension overload	4.5.2.10.3	3.11.21
X			Stability	4.5.2.11	3.11.17
X			Frame racking test	4.5.2.12	3.8.1
X			Overhead safety guard	4.5.2.13	3.11.18
X			Operator's platform	4.5.2.14	3.8.2.1
X		X	Service brake	4.5.2.15	3.11.8
X			Power train	4.5.2.16	
X			Electromagnetic interference	4.5.2.17	3.16
X			Power consumption	4.5.2.18	3.11.19
X			Endurance	4.5.2.19	3.11.15
X		X	Battery removal	4.5.2.20	3.7.2.2

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4.5.2.1 Lift and tilt tests.

4.5.2.1.1 Fork height. Elevate rated load to maximum lift height with load extended. Measure the distance from the ground to the top face of the forks with load horizontal. The fork height shall not be less than specified in 3.11.11.

4.5.2.1.2 Free lift height. Elevate rated load (forks retracted) until the inner mast, fork carriage or forks begins to exceed the specified collapsed mast height. Measure the true vertical distance from floor to top surface of the forks at rated load center. The free lift height shall be as specified in 3.11.12.

4.5.2.1.3 Fork tilt. The truck with rated load in carrying position (forks retracted) shall be placed on a level horizontal surface. With forks parallel to the floor, mark a vertical reference line on the side of load. Tilt load to maximum rearward position and measure angle of travel by referenced line. The load shall then be raised to maximum lift height, fully extended and maximum rearward tilt and maximum forward tilt. A clinometer, protractor or other measuring device may be used for this test. Tilt as measured shall not be less than specified in 3.11.9.

4.5.2.1.4 Lifting speed and lowering speed. Starting with a fully charged battery and hydraulic oil at, ambient temperature, the empty forks, fully retracted, shall be raised to maximum height. Actuate lower control and record time required for unloaded forks to reach their fully lowered position. Repeat this operation ten times, calculate average time, and compute lowering speed. Lowering speed shall be as specified in 3.11.2. Lowering speed shall also be checked under simulated hydraulic line failure to determine compliance with 3.7.4. After computation of lowering speed, place rated load on the forks with center of gravity of the load positioned 24 inches from the front face of the forks. Raise load at maximum lifting speed to full lift height and record the time required to reach this height. Repeat lift cycle five times and calculate average time. Repeat test without load. Speed of lift with and without rated load shall not be less than specified in 3.11.1.

4.5.2.2 Extension and retraction. Raise rated load to maximum lift height in five (5) interval steps. Extend and retract rated load at each step to the full horizontal distance. Measure this distance and record the time in seconds from the instant extension or retraction begins to the end of extension or retraction to determine compliance with 3.11.3.

4.5.2.3 Underclearance test. The truck shall be operated from one horizontal plane up a ramp to another horizontal plane and down the ramp without rated load. Observe whether all parts of the truck clear the surface. The truck, with rated load on the forks, shall be operated from one horizontal plane up a ramp to another horizontal plane and down the ramp to determine compliance with the ramps specified in 3.11.13.

4.5.2.4 Steering tests.

4.5.2.4.1 Steering gear. The steering mechanism shall be tested to determine that the operator can comfortably turn the wheel, without

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interference from adjacent truck structure, from extreme left to extreme right and vice versa, within limits specified in 3.11.4. Forces shall be applied to the steering knob in the plane of the wheel. Test shall be conducted with truck stationary on dry level concrete.

4.5.2.4.2 Steering overload. The truck shall be tested to determine compliance with the steering gear assembly overload requirements of 3.11.4. Loads shall be applied in the plane of the steering wheel. The steering wheel shall be turned to full right and full left to determine compliance with this requirement. The test shall be conducted 5 times in each direction alternately with the steering mechanism at the extreme turn of the steering wheel to the right and to the left.

4.5.2.4.3 Free play. The steering mechanism shall be inspected to determine compliance with 3.11.15. Measurements shall be taken before and after endurance testing.

4.5.2.5 Speed. A measured distance of 44 feet (13.4m) shall be laid out over a dry, smooth, level, concrete surface. The beginning and end of the test course shall be clearly identified. The loaded truck shall be operated a sufficient distance to attain maximum speed prior to entering the measured test course. With a timing device, record the time required for truck to travel the measured distance. Drive truck in the reverse direction and record travel time over the same measured distance. This procedure shall be repeated at least 3 times. Alternate methods for checking speed may be used, such as electric eyes, electronic or other accurate means, and when used, may be timed through shorter distance. The time required for travel in each direction of travel shall be separately averaged. The averaged time in each direction shall not exceed 7.5 seconds. Test is to be conducted with a fully charged battery and with tires warmed-up by 10 or 15 minutes of running.

4.5.2.6 Acceleration. A measured distance of 22 feet (7m) shall be laid out over a dry, smooth, level, concrete surface. A reference point shall be marked on the truck so as to be readily seen by the operator and test personnel while truck is moving. The truck with rated load shall be positioned so that the reference mark on the truck is directly over the starting mark. The truck shall be accelerated as in normal operation and the time recorded when the truck reference mark moves from beginning to end of test distance. This shall be repeated not less than three times in both the forward and reverse directions. The time required to travel in each direction shall be separately averaged. Time for travel in each direction shall not exceed the time specified in 3.11.7.

4.5.2.7 Slope ascension, forward direction. The truck with no load shall be driven onto a 15 foot (4.6m) long ramp having a 5 percent slope. Truck shall be brought to a complete stop when all wheels are positioned on the ramp. Accelerate truck from this location up the ramp to a horizontal level. Test shall be conducted with a fully charged battery.

4.5.2.8 Maneuverability. The test shall be conducted in an aisle width as applicable (see 3.11.16), with side panels or enclosures at least 5 feet (1.5m) high and a stacking base for load placement not less than 7 feet (2.1m) high. The stacking base shall be framed or enclosed with side panels to simulate a typical storage rack 56 inches (1420mm) wide for style I trucks and 54 inches

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(1370mm) wide for style II trucks. Test load shall be a 40-inch (1020mm) long by 48-inch (1220mm) wide pallet with rated load. Time measurements for each placement, or removal cycle shall begin with the truck parallel and within the aisle with forks approximately 8 inches (200mm) above the floor in the retracted position, and the cycle shall end with the truck in approximately the same position (forks empty or loaded) in the lowered position. Time measurements shall include the time required to maneuver, to stack, lifting load to stack height, extending load, placement on the stacking base, return with forks empty, and lowering forks to 8 inches lift height. Similarly, the reverse cycle shall include lifting empty forks to the height of the pallet on the stacking base, maneuvering into position, engaging the pallet, raising the load off the stacking base, retracting load, returning to the original position, and lowering load to carrying position. The operation shall be perfomed at least 9 out of 10 times, without reversal of direction, with not more than 60 seconds being required for each placement or removal cycle. This testing may be conducted by contractor's selected operator, utilizing such aids and skillful manipulation as may be required to provide the best mechanical advantage to the opeator and the equipment. Operation may include, but not be limited to: (1) maneuvering with load raised to an efficient height, but not higher than the stacking base for the pallet; (2) operation with load in rearward tilt position; (3) in the removal of a pallet from stack position, entry of the forks into the pallet opening prior to attaining a position for the truck in which its longitudinal axis is fully perpendicular to the face line of the pallet stack; (4) similarly, but conversely, in the removal of the truck forks from a pallet which has been placed in position, truck motion may follow any preferred pattern of movement to permit most efficient withdrawal possible without damaging adjacent or opposite pallet stack or barriers. In placement and load removal cycles, the pallet and load support on the forks shall be with any vertical face or side at 90 degrees to either the longitudinal or lateral axis. The operator may use any other recognized techniques or manipulation commonly practiced in commercial or military handling operations, with the exception of direct reversal during any one placement or removal action.

4.5.2.9 Load back rest. A static thrust load equivalent to rated load of the truck shall be applied to the fork rack uniformly distributed over the entire vertical surface from the load supporting surface of the forks (forks retracted) to the extreme top of the load backrest without causing fracture at the welded joints, deformation of any frame member or failure in the rack mounting.

4.5.2.10 Lift assembly.

4.5.2.10.1 Shock. The lift mechanism shall be tested for effectiveness to absorb shock and prevent overturning of the truck and damage to lift system, when rated load is lowered at maximum speed in retracted position and fully actuated control is released abruptly. Repeat test 20 times. Examine truck to determine compliance with specified requirements, deformation, damage, or leakage of fluids which constitutes failure of the test.

4.5.2.10.2 Retraction Overload. With pressure relief valve inoperative, raise retracted forks to a minimum of 30 inches (760mm) lift height and place 300 percent of rated load on forks and retain for ten minutes. Remove load and examine truck for evidence of external leaks or breakage of the hydraulic

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system as a result of this test. The load support assembly, frame, and related structure shall be examined for permanent deformation or any indication of structural failure.

4.5.2.10.3 Extension overload. Extend load support assembly to full forward position, with forks raised at least 12 inches (305mm) above the floor and rear of truck counterweighted or tied down. Place 200 percent rated load on the fork assembly and retain for 5 minutes. Remove load and examine the truck for evidence of failures. The pressure relief valve shall be made inoperative for this test.

4.5.2.11 Stability. Tests to determine compliance with the stability requirements of 3.11.17 shall be conducted in accordance with ANSI B56.1.

4.5.2.12 Frame racking test. Determine maximum static force for each support under loaded and unloaded conditions with forks extended and retracted. Secure all but one support area of the truck in a jig or similar device, and apply a vertical force equal to 200 percent of the static force determined for each support area. Complete this test by applying load first at one and then the other of two adjacent corners. Examine frame and related structure for permanent deformation or distortion as a result of this test.

4.5.2.13 Overhead safety guard. The overhead safety guard shall be tested in accordance with ANSI B56.1. Nonconformance to the requirements of 3.11.18 shall constitute failure of this test.

4.5.2.14 Operator's platform. With the truck anchored to prevent movement apply a force equal to the loaded weight of the truck. The load shall be applied to an area 3.5 inches (90mm) high and 3 inches (76mm) wide maximum with its center 42 inches(1067mm) above the floor. The load shall be applied at the right hand rear corner of the compartment. Any evidence of failure or permanent deformation shall constitute failure of this test.

4.5.2.15 Service brake test. Service brake test shall conform to ANSI B56.1.

4.5.2.16 Power train static test. The static torque test shall be accomplished by using a torque arm with one end attached to the motor shaft and with the other end attached to a dynamometer. Drive wheel(s) shall be locked to prevent any movement. The force shall be applied through the dynamometer to the torque arm by use of chainfall or similar device. Applied torque shall be equivalent to 150 percent of the maximum torque capable of being developed by the traction motor. At the completion of the test, disassemble and inspect the power train for possible failure due to external and internal stresses. This test may be accomplished on the entire power train assembly prior to assembly of the truck, at the option of the contractor.

4.5.2.17 Electromagnetic interference. The truck shall be tested with all motors operating as specified in SAE 551c.

4.5.2.18 Power consumption. The truck, equipped with a fully charged battery as specified in 3.7.2.1, shall be operated continuously over the test course shown on figure 1 with the same procedure and test loads described in paragraph 4.5.2.19. A single fully charged battery shall be used for the

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duration of this test. The truck shall not be stopped except to alternate operators if necessary. Each test cycle shall be timed and recorded. The test shall be conducted as part of endurance test 4.5.2.19 except that it shall constitute the first five hours of the endurance test. Cycles shall be repeated continuously for a minimum of five continuous hours without lift cut-out by battery protector described in paragraph 3.7.2.3 or leakage of fluid, inability to perform any function or malfunction of any kind shall contitute failure of this test.

4.5.2.19 Endurance. The truck shall be subjected to 2,000 total (including power consumption) complete circuits of the test course shown in figure 1. The test shall be conducted on a continuous basis during each operating 8 hour day. Battery changes shall be permitted as necesary to assure continuous operation for each operating day. At the completion of each 1,000 circuits of the course, manufacturer's recommended preventive maintenance shall be performed and the truck inspected to assure all components are functioning properly. No repair or replacement of parts shall be permitted. The course shall be negotiated one-half the total circuits in a clockwise direction and one-half in a counterclockwise direction as follows. The course need not be rearranged for the counterclockwise operation. Rated loads plus zero, minus 30 pounds (14kg), shall be placed on standard 40-inch (1020mm) long by 48-inch (1220mm) wide pallets. From point "A" proceed to point "B" and remove load from platform. Retract load fully and proceed rearward to point "C" with load at normal carry height. Deposit load on the platform at point "C", retract forks and proceed to point "D". Pick up load at point "D" and proceed rearward to starting point "A". At point "A" stop the truck, extend the forks and lower to ground level, blow horn, actuate light switch, and fully apply brake. Raise and retract forks to carry position. Proceed forward to "B" and deposit load. Continue to negotiate the course in this manner, alternately placing and retrieving loads until the required number of test circuits has been completed. All turns into and out of the stack areas shall be smooth and continuous. Inability to negotiate the course for 2,000 complete cycles, evidence of excessive wear, leakage of fluid, inability to perform any function or malfunction of any kind, or failure of any component shall constitute failure of this test.

4.5.2.20 Battery removal. Measure force to maintain movement of battery to determine conformance to requirement of 3.7.2.2.

4.6 Post-test inspection. Upon completion of tests in 4.5.2, the truck shall be subjected to the tests marked X in column 3 of Table IV. Nonconformance with applicable requirements shall constitute failure.

4.7 Inspection of preparation for delivery. The trucks shall be examined to determine conformance to the preservation, packaging, packing and marking requirements of MIL-STD-162.

5. PACKAGING

5.1 Preservation, packaging, and packing. Preservation, packaging and packing shall in accordance with MIL-STD-162. (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, (see 6.2). Marking shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Trucks described herein are intended primarily for use inside warehouses handling palletized cargo where narrow aisle widths and low floor load ratings prevent the use of less maneuverable and heavier trucks. Trucks are intended for operation over paved surfaces.

6.2 Acquisition Requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.

(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

(c) Type, style and size (including capacity) required. (See 1.2).

(d) When first article test is not required. (See 3.2).

(e) When cold storage conditioning is required. (See 3.4).

(f) When fork dimensions are to be other than shown in Table I. (See 3.8.3.1).

(g) When frame racking test is not requisred (See 3.8.1).

(h) When remote control is required. (See 3.9.1.5).

(i) When finish color is other than specified. (See 3.14).

(j) Preservation, packaging and packing requirements. (See 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data is requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To insure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) shall be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.3	DI-T-4902	First Article Inspection Test Report	_____

The above DID was cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management System and Data Requirements Control List (AMSDL), must be researched to insure that only current, cleared DID's are cited on the DD Form 1423.

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6.4 Technical Manuals. The requirements for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a first article sample, or a standard production item from the contractor's current inventory (see 3.2) and the number of items to be tested as specified in 4.3. The contracting officer shall also include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results, and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Conditions for use of level B preservation. When level B preservation is specified (see 5.1), this level of protection should be reserved for the acquisition of forklift trucks for resupply worldwide under known favorable handling, transportation and storage conditions.

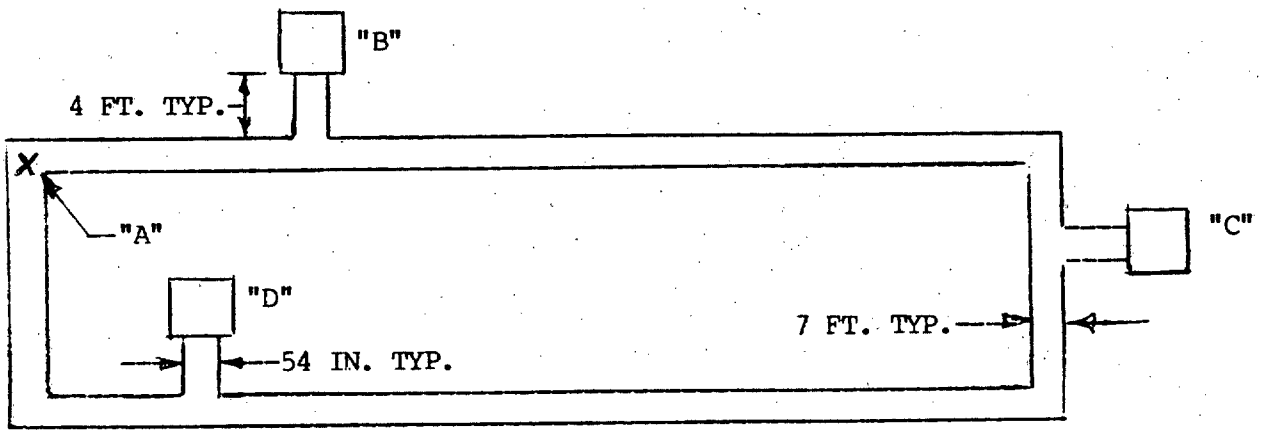
6.7 Subject term (key word) listing.

Electric
Fork
Lift
Truck
Reaching
Tiering
  Narrow

6.8 Changes from previous issue. Marginal notes are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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POINT	STACK HEIGHT
B	Specified fork height
C	Half specified fork height
D	FLOOR



TOTAL COURSE LENGTH - 320 FT. APPROX.

FIGURE 1. ENDURANCE COURSE

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Custodians:

Army - ME
Navy - SA
Air Force - 99

Review Activities

Army - MT, AT
Navy - YD
DLA - CS

Preparing activity:

Navy - SA

Project 3930-0641

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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